## a Ingestion and biophysical gate

**VIIRS 375 m**
Suomi-NPP · NOAA-20/21

**ERA5-Land**
3-hourly → daily · 0.25°

**MapBiomas C2**
annual · prior-year join

**Hansen GFC**
tree-cover loss

**Historical priors**
causal · 3-year window

*daily 1-km² stack* →

**Biophysical gate**

Fuel = 1

VPD ≥ 0.8 kPa

$R_{48h} < 10$ mm

excludes non-combustible and wet cell-days

↓

**Gated candidate cell-days**

21.7% cell-days removed

## b D2 target operator

$F_t = \{i : \text{fire}(i,t) = 1\}$

$I_{i,t} = 1 \Leftrightarrow i \in F_t,$ $I \cap P = \emptyset$

$N_{25}(i, t-1) = \emptyset,\ H_{10}(i,t) = 0$

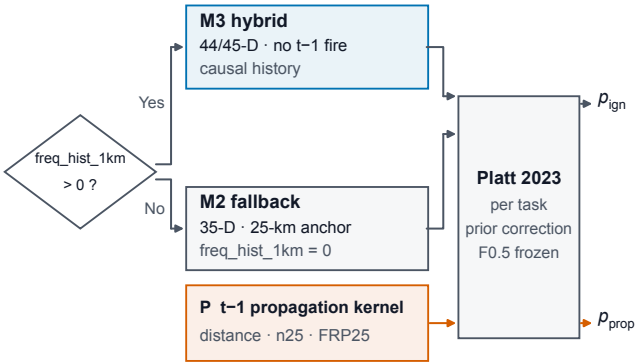
$P_{i,t} = 1 \Leftrightarrow i \in F_t,$ 25 km · 10 d

$N_{25}(i, t-1) \neq \emptyset$

**I primary ignition** | **P tactical propagation**

*domains disjoint*

## c Decoupled inference

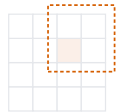



## d Auditable evaluation

**Temporal split** 1:10 case–control

| 2018–2022 | 2023 | 2024–2025 |
|---|---|---|
| Train | Val | Test frozen |

**Spatial holdout**



50-km blocks
20-km buffer
*spatial only*

**Uncertainty**

date-grouped
bootstrap
500 CI · 1,000 paired

**Decision value**

cost-loss

**Open code + data**

reproducible release